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Roll No. 3112413012

324732(24)

BE (7th Semester)
Examination, Nov.-Dec., 2016

Modern Control System

Time Allowed : 3 hours *Maximum Marks :* 80
Minimum Pass Marks : 28

Note : (*i*) All questions are compulsory. Part (*a*) of each question is compulsory. Do any two from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) What are describing functions? [2]

(*b*) Derive the describing function of relay with dead zone and hysteresis whose input-output characteristic is as shown in Figure 1. [7]

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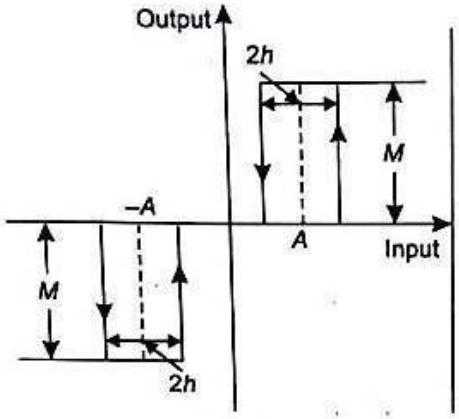

Figure 1

(*c*) What are limit cycles? Determine the amplitude and frequency of the limit cycle of the system shown in the Figure 2 with input *x* as sinusoidal. [7]

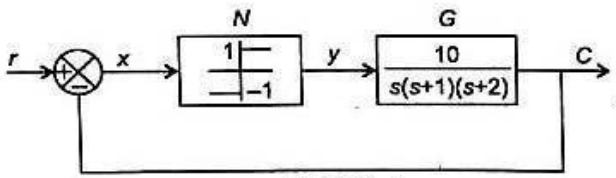

Figure 2

(*d*) What are singular points? Consider a second order system described by

$$\frac{d^2y}{dt^2} + 2\zeta\omega_n \frac{dy}{dt} + \omega_n^2 y = 0$$

According to the values of the characteristic roots, find the singular points and give the names of these points. [7]

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2. (*a*) Prove the non-uniqueness of the state models. [2]

(*b*) Explain the conversion of transfer function to state variable models in Controllable Canonical Form (CCF), Observable Canonical form (OCF) and Jordan canonical Form. Obtain the three different realizations for the transfer function

$$G(s)=\frac{Y(s)}{U(s)}=\frac{s+3}{s^3+9s^2+24s+20}$$ [7]

(*c*) Prove that for

$$A=\begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 & 0 \\ -a_2 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ -a_{n-1} & 0 & 0 & \cdots & 0 & 1 \\ -a_n & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

with distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, the diagonalizing matrix is given by

$$M=\begin{bmatrix} 1 & \cdots & 1 \\ \lambda_1+a_1 & \cdots & \lambda_n+a_1 \\ \lambda_1^2+a_1\lambda_1+a_2 & \cdots & \lambda_n^2+a_1\lambda_n+a_2 \\ \vdots & & \vdots \\ \lambda_1^{n-1}+a_1\lambda_1^{n-2}+\cdots+a_{n-2}\lambda_1+a_{n-1} & \cdots & \lambda_1^{n-1}+a_1\lambda_1^{n-2}+\cdots+a_{n-2}\lambda_1+a_{n-1} \end{bmatrix}$$

[7]

(*d*) Show that if there is pole-zero cancellation in the input-output transfer function of a linear time-invariant system, the system will be either uncontrollable or unobservable. [7]

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3. (*a*) What is Liapunov function? [2]

(*b*) State and explain the four theorems to investigate stability using Liapunov's method. [7]

(*c*) Explain the variable gradient method to formulate the Liapunov function from which the stability of the equilibrium state may be determined.

Check the stability of the system described by

$$\dot{x}_1=-x_1$$
$$\dot{x}_2=-x_2+x_1x_2^2$$

by use of variable gradient method. [7]

(*d*) State and prove Liapunov's theorem of stability for linear systems. Determine the stability using the Liapunov method for a system represented by

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}=\begin{bmatrix}-1 & -2\\ 1 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$ [7]

4. (*a*) Draw the respective block diagrams showing the state feedback control schemes when all the state variables can be measured, and when some of them have to be estimated. [2]

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(*b*) Consider the state equation $\dot{x} = Ax + Bu$ where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & -10 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}$$

The following design specifications are to be met $\zeta = 0.5$, $\omega_n = 2$ rad/sec. Choose the third pole at $s = -10\ \zeta\omega_n$. Determine the necessary feedback gain matrix K and the control signal u. [7]

(*c*) Explain the terms full order and minimum order state observers. For a state model with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, \ B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \ C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

Assuming that the output y can be accurately measured, design a reduced order observer with the desired eigenvalues for the reduced order observer as

$\mu_1 = -2 + j3.464, \quad \mu_2 = -2 + j3.464,$ [7]

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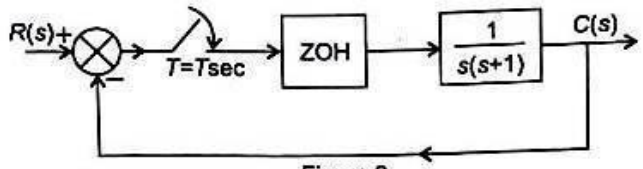

Figure 3

(*d*) Derive Ackermann's formula for the determination of the state feedback gain matrix K and state observer gain matrix K_e [7]

5. (*a*) Obtain the pulse transfer function when the input output relation is given in the form of difference equation. [2]

(*b*) What is sampling and zero-order hold (ZOH)? What is the transfer function of ZOH. For the sampled data control system shown in Figure 3, find the output $C(z)$ when the input is a unit step input taking the sampling period $T = 1$ sec. [7]

(*c*) Determine the stability of the following characteristic equations using Jury's test :

(*i*) $Q(z) = z^3 + z^2 + 0.2z + 0.5$

(*ii*) $Q(z) = z^3 + z^2 + 0.2z - 0.5$

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(*d*) For the system shown in Figure 4, compute the value of K that minimizes the following performance index for the unit step input.

$$J = \int_0^\infty \left[e^2(t) + \lambda u^2(t) \right] dt; \lambda = 0.5$$

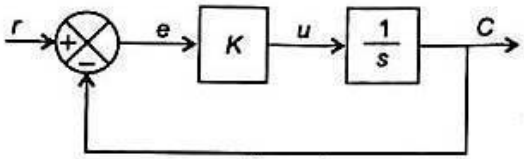

Figure 4

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